ATrack

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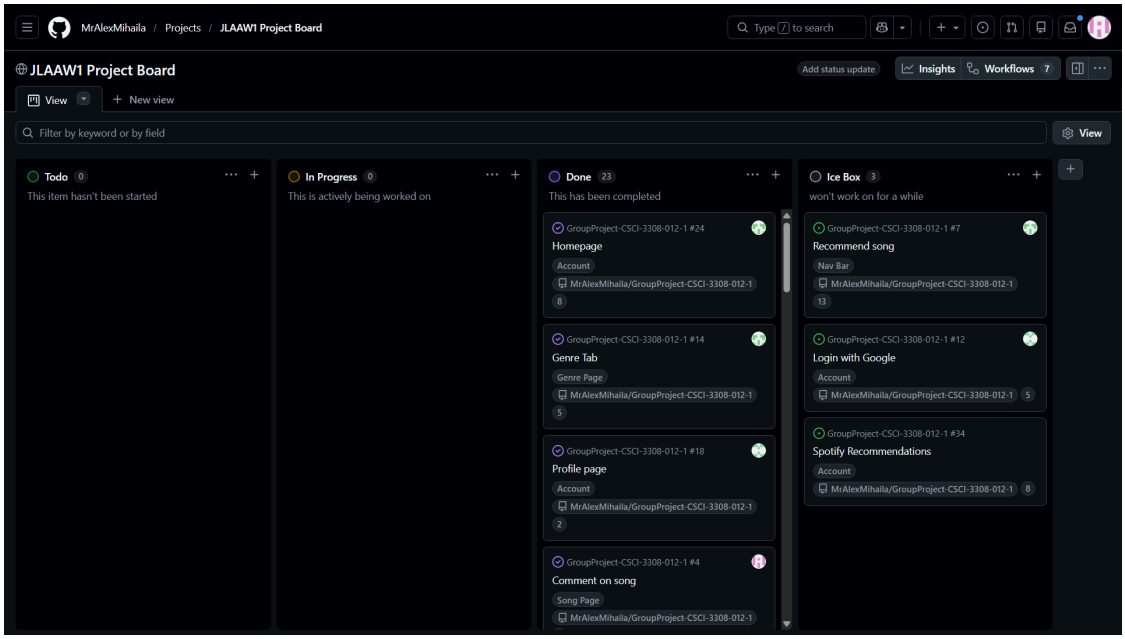
William Brito (WillisBurr)

Project Description

ATrack is a music review platform that allows users to publicly grade a song, album, or single from A-F, as well as leave an optional written description explaining their reasons for said rating. On the main home page, users can browse popular songs, genres, and albums, as well as view activity from their friends. A search feature allows users to find any artist, album, song, or user on the platform. Users can also view, like, and dislike comments made by other users. What sets ATrack apart from other music review applications like MusicBoard is its integrated timestamp comment system. Users who connect their Spotify Premium account can listen to a song directly on the page and leave comments at specific timestamps in the songs, allowing users to share their opinions during specific moments in a track, not just the song as a whole. ATrack also cultivates a social experience. By allowing users to follow friends and other music enthusiasts, ATrack makes it easy to keep up with reviews and music activity in real time. This social approach helps to encourage music discovery and engagement like never before.

Project Tracker

Link: https://github.com/users/MrAlexMihaila/projects/1



Demo Video

https://drive.google.com/file/d/1WLmWlzC8mVEQyk8-XopQrIcqlYnSLFDB/view?usp=sharing

Version Control System

Link: https://github.com/MrAlexMihaila/GroupProject-CSCI-3308-012-1

Contributions

Luke McCord

Implemented the search feature allowing the user to search for any artist/album/song using the Spotify API. Built the songs and albums page to each display a list of the most popular songs and albums respectively. Also developed the individual artist page that populates with the artist's most popular songs and all of the artist's albums. Implemented dark mode, giving users a second appearance to use. Technologies used: JavaScript, HTML, Handlebars, CSS

Jonathan Stevenson

Designed and diagramed the schema for the database to store songs, albums, artists, genres, timestamped comments, and reviews for both songs and albums. Implemented the profile page to allow the user to view their recent reviews, friend count, follower count, and profile picture. Additionally implemented profile picture functionality using Multer to handle uploads for files that are temporarily stored in the server, then uploaded to Cloudinary for cloud image hosting. Created a menu in the navigation bar to house profile, logout, and spotify connection functionality, allowing for a less cluttered sidebar. Technologies used: JavaScript, HTML/Handlebars, CSS, Cloudinary, Multer

Andrew Fong

Worked on creation and design of wireframes for implementing user test cases and achieving user acceptance criteria. Implemented friends page to populate based on search. Built the genres page to display top artists and top songs in that genre based on the spotify API. Built a home page to display the top songs and top albums according to the letter average grading system based on reviews on songs and albums from left to right. Additionally implemented the friends feed to display the last five reviews from users follows on songs and albums showing when its creation, grading, and message also allows for a redirect to that song/album. Technologies used: JavaScript, HTML/Handlebars, CSS

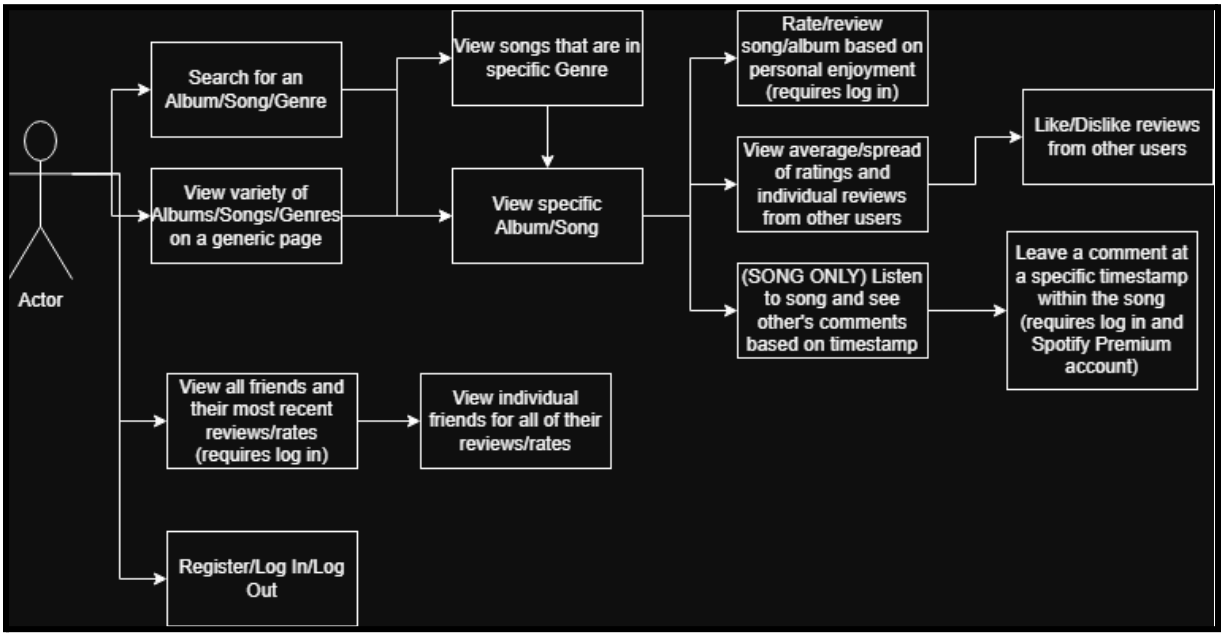
Alex Mihaila

Did the initial setup of the project. Implemented the Spotify Web API connection for others to use. Implemented all features of the individual song page, including rating the song on a scale from A to F, viewing ratings from other users, like/disliking user ratings, a song player using the Spotify Web Playback SDK that requires a connected Spotify Premium account, adding a comment at a specific timestamp in a song, and viewing other user's timestamp comments. Additionally assisted in improving the user interface of the website. Technologies used: JavaScript, HTML/Handlebars, CSS

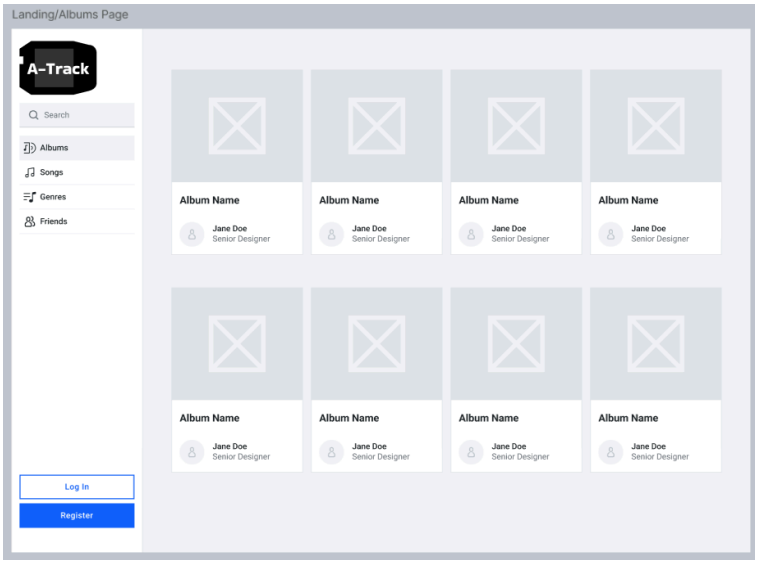
William Brito

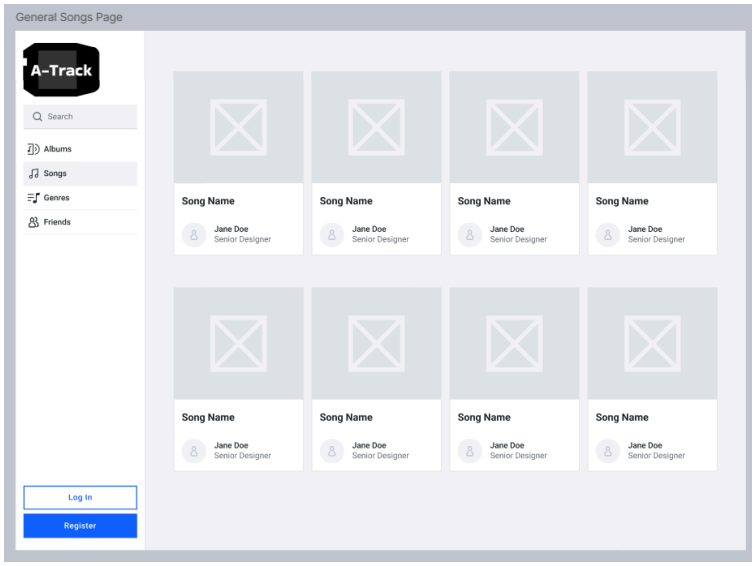
Handled basic sign-in/registration to an account and all features of the individual album page. The implemented features include login/registration, giving all pages access to the logged in user's token, displaying details about an individual album, allowing users to rate the album from the scale of A to F, displaying other user's ratings on an individual album, and displaying all songs in an album with their ratings. Technologies used: Javascript, HTML/Handlebars

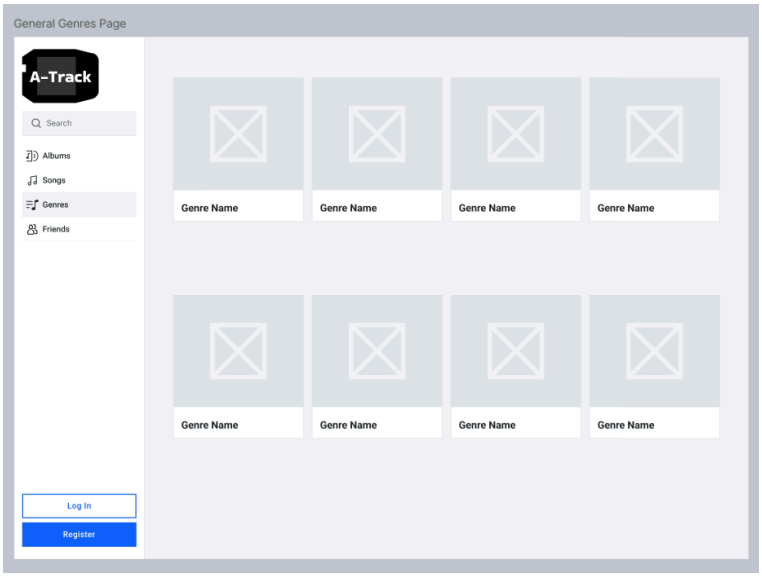
Use Case Diagram

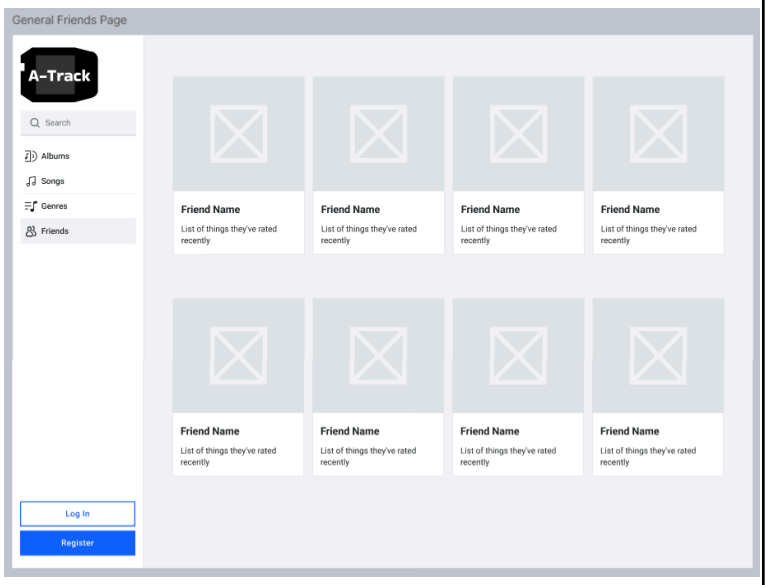


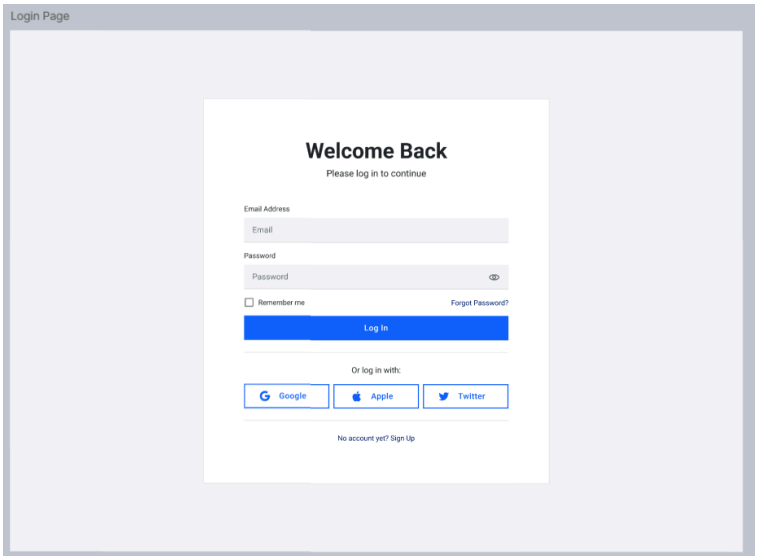
Wireframes

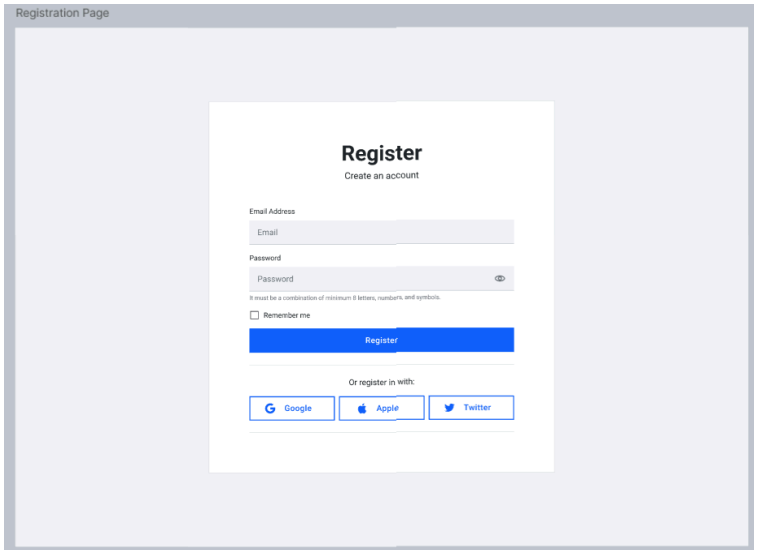

Landing/Albums Page
A-Track
Search
Albums
Songs
Genres
Friends
Log In
Register
Album Name
Jane Doe
Senior Designer

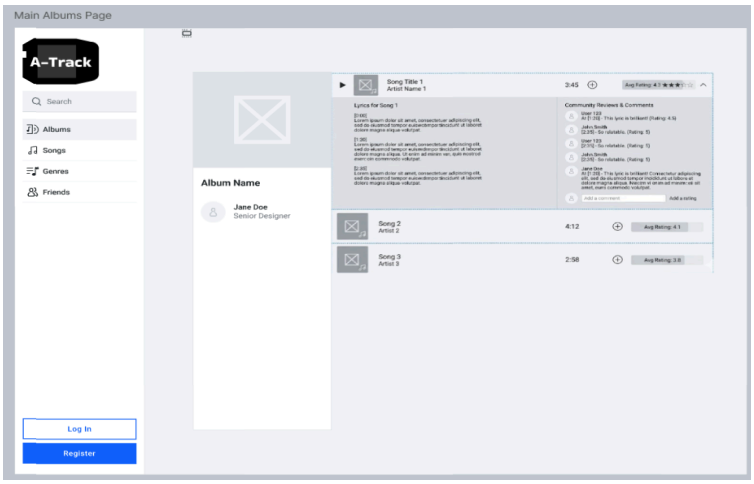

General Songs Page
A-Track
Search
Albums
Songs
Genres
Friends
Log In
Register
Song Name
Jane Doe
Senior Designer

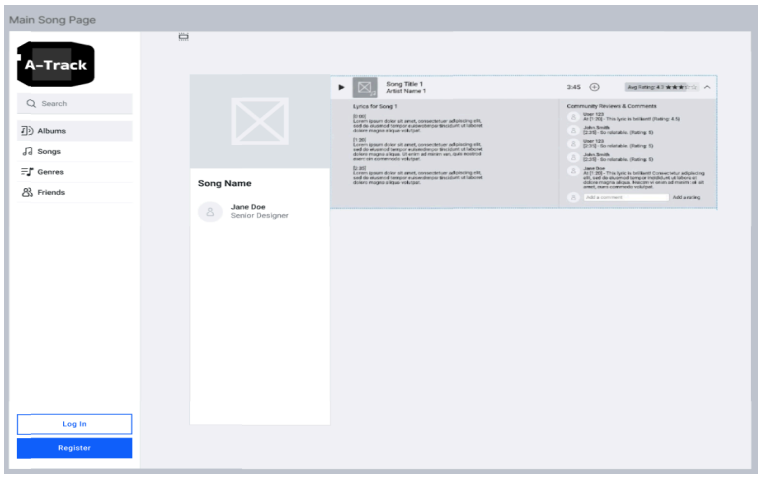

General Genres Page
A-Track
Search
Albums
Songs
Genres
Friends
Log In
Register
Genre Name

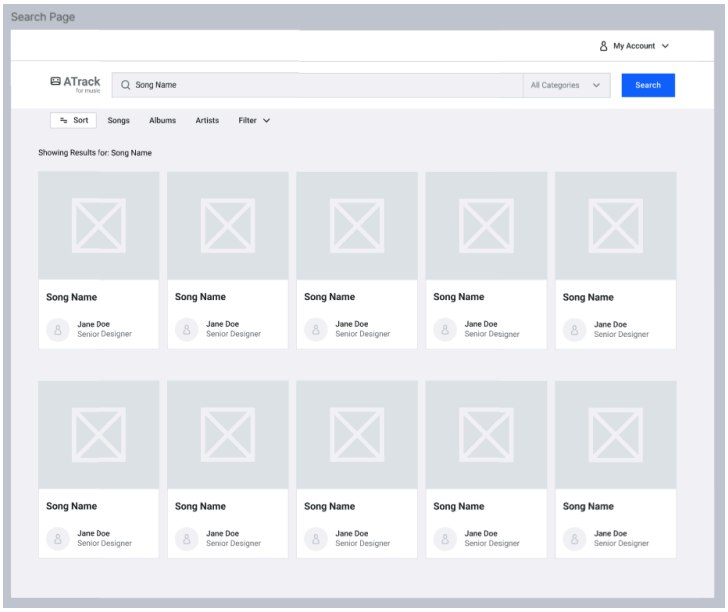

General Friends Page
A-Track
Search
Albums
Songs
Genres
Friends
Log In
Register
Friend Name
List of things they've rated recently


Login Page
Welcome Back
Please log in to continue
Email Address
Email
Password
Password
Remember me
Forgot Password?
Log In
Or log in with:
Google
Apple
Twitter
No account yet? Sign Up


Registration Page
Register
Create an account
Email Address
Email
Password
Password
It must be a combination of minimum 8 letters, numbers, and symbols.
Remember me
Register
Or register in with:
Google
Apple
Twitter


Main Albums Page
A-Track
Search
Albums
Songs
Genres
Friends
Log In
Register
Album Name
Jane Doe
Senior Designer
Song Title 1
Artist Name 1
3:45
Lyrics for Song 1
Community Reviews & Comments
Add a comment
Add a rating
Song 2
Artist 2
4:12
Avg Rating: 4.1
Song 3
Artist 3
2:58
Avg Rating: 3.8


Main Song Page
A-Track
Search
Albums
Songs
Genres
Friends
Log In
Register
Song Name
Jane Doe
Senior Designer
Song Title 1
Artist Name 1
3:45
Lyrics for Song 1
Community Reviews & Comments
Add a comment
Add a rating



Test Results

Feature 1: User Log-in / Authentication
This test case evaluates the application's ability to authenticate users securely and accurately. Testers attempted to log in using a combination of valid credentials, invalid passwords, and empty fields to verify that the system correctly validates input against the database. The goal was to confirm that valid users are redirected to the home page, invalid attempts produce clear error messages, and user sessions persist after a successful login.

<u>Observations:</u>
Testers navigated to the login page and filled in the username and password fields before clicking the Login button, which aligned closely with the expected use case. Behavior was mostly consistent across testers, and the website's behavior was predictable and expected. Valid credentials successfully redirected, and invalid or empty submissions triggered appropriate error messages as designed. One deviation occurred when a tester submitted the form using the Enter key rather than clicking the button, which deviates slightly from the test plan but still triggers form submission. The application met all acceptance criteria and no changes were required for this feature.

Feature 2: Commenting on Song
This test case verifies that authenticated users can successfully submit a comment tied to a specific timestamp on a song, and that the comment is stored and visually displayed on the song page at the correct position. Testers were required to have an active session and were

asked to complete the comment form using both valid and incomplete data to confirm that the timestamp and description fields are both enforced as required. The expected outcome was that valid submissions appear on the song page after submission, and invalid submissions were not saved.

Observations:
Testers were able to search for a song and navigate to the song page without difficulty. Once on the song page, testers found the comment form without difficulty, but several users experienced difficulty with changing the timestamp they wanted to review. The timestamp text box is read-only, so users must change the timestamp from the playback slider, which took a bit of time to figure out for some. We did not make a change to the application because we did not want to allow users to input an incorrectly formatted timestamp. The application handled empty and invalid timestamp comments correctly, not allowing those to be entered into the database or displayed on the song page. This feature meets all criteria for user acceptance testing.

Feature 3: Searching for a specific song
This test case evaluates the application's ability to accurately retrieve songs based on user search input. Testers entered a variety of search queries, such as song titles, incorrect or partial matches, and empty inputs to determine if the search feature works as intended. The goal was to confirm that valid searches returned a relevant populated results list including relevant metadata about said song, and that empty submissions return a message notifying the user that the field cannot be empty.

Observations:
Testers entered queries in the search bar with ease, and behavior was consistent across all testers. The application responded predictably to both valid and empty search queries. Valid search requests returned a list populated with all of the top results for that song with corresponding metadata including album art. Empty search requests were appropriately blocked and returned a message to the user saying "Please fill out this field." Searches that return no matches display a results page with the appropriate "No results found for this search." The application met all the criteria for user acceptance testing, and no changes to the application were required.

Feature 4: Following another user
This test case verifies that a user, when logged in, is able to go to either the friends tab or search for users, select a profile of a user based on the results, and then choose to follow that user. Testers had the option to either use the friends tab or to search a user using the search feature. The goal was to confirm that following other users worked and that the process was seamless for users.

Observations:
Testers were able to find profiles with ease and follow other users. Behavior was consistent across all testers. The application responded as expected. Testers did not have any issues with

the user interface or flow in order to follow another user. The application met all the criteria for user acceptance testing, and no changes to the application were required.

Deployment

Link: https://groupproject-csci-3308-012-1.onrender.com/home
The application was deployed using Render. To access, one must simply open the link above.

Creativity of Website from UI/UX

i. Cohesive Visual Design:

ATrack maintains a consistent visual identity across all pages through a unified color palette (white/black with purple accents), as well as a standardized album card and song card to ensure every result shows up uniformly. Our color palette is intentionally minimal, so that the songs/album art remains the primary focus. This consistency throughout the website makes it a pleasant and easy experience for the user.

ii. Clear Flow and Intuitive Navigation:

Navigation is designed to be incredibly intuitive. The navbar on the left side of the site provides clear, concise labels for navigation to different pages such as albums, songs, genres, friends. Also contained in the navbar is a search bar which makes it easy to find your favorite tracks or artists or albums. Our layout also helps to prioritize discovery and exploration through components like horizontal scroll bars so the users can scroll through categories like “Today’s Top Hits.”

iii. Custom Styling Beyond Minor CSS Tweaks:

We utilized CSS for a lot more than just minor tweaks. For example, on our search or discovery pages, we used CSS to add a hover effect to the album/song/artist cards to help the experience feel more interactive. In addition to this we used CSS to make layouts for results pages, truncating text if it was too long, as well as implementing dark mode. Additionally, custom CSS is used to control layout structure across most pages, including grid sizing, responsive card layouts, and horizontal scroll bars.

iv. Engaging UI Elements (Animations, Branding, Dynamic Feedback):

ATrack’s interface includes many interactive and visually engaging elements that work together to enhance the user experience. At the top of the navbar there is a custom logo, designed by Jonathan Stevenson which helps to establish a distinct identity. In addition to this, the previously mentioned hover effects on artist/album/song cards. These hover effects help to immerse the user in the app, and make it more of an experience than a website. It shows to the user that their control of the mouse is actually changing things in the website. It also helps to highlight what the user is hovering over, which can sometimes be difficult.

v. Accessibility Considerations:

Accessibility is always incredibly important when it comes to UI design. A key accessibility feature in ATrack is the implementation of a dark mode toggle. When a user presses the crescent moon button on the left of the page, the site will toggle to dark mode. Dark mode is incredibly important for users in low light situations, may be sensitive to the bright white

screen, and can reduce visual strain. We ensured our text was always readable by prioritizing contrast, white text on black background, black text on white background, etc. In addition to this, hover effects can be considered an accessibility feature. One of the most important aspects of designing a UI is always knowing where your mouse is. One way to make it easier to tell where it is, is to add hover effects. This makes it so that if your cursor is on a card, it will “hover” off the rest of the page.